SeeWandel Climate

Interreg
Alpenrhein-Bodensee-Hochrhein

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# Regime Shifts in Plankton Dynamics: What Drives Them and Can We Predict Them?

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www.seewandel.org

Ecomeets at Eawag, Dübendorf
28 October 2026 · 11:00–12:00

# Slides and links

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SORTEE
Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology

Slides, code and data shared openly
leomarameo7.github.io/science_communication.html

# The Pomati lab




**Plankton Ecology Group**
PI Dr. Francesco Pomati, Eawag

**SeeWandel-Climate**
Lake Constance plankton and fish under climate change

Co-funded by the European Union
(Interreg Alpenrhein-Bodensee-Hochrhein)

# Take home message

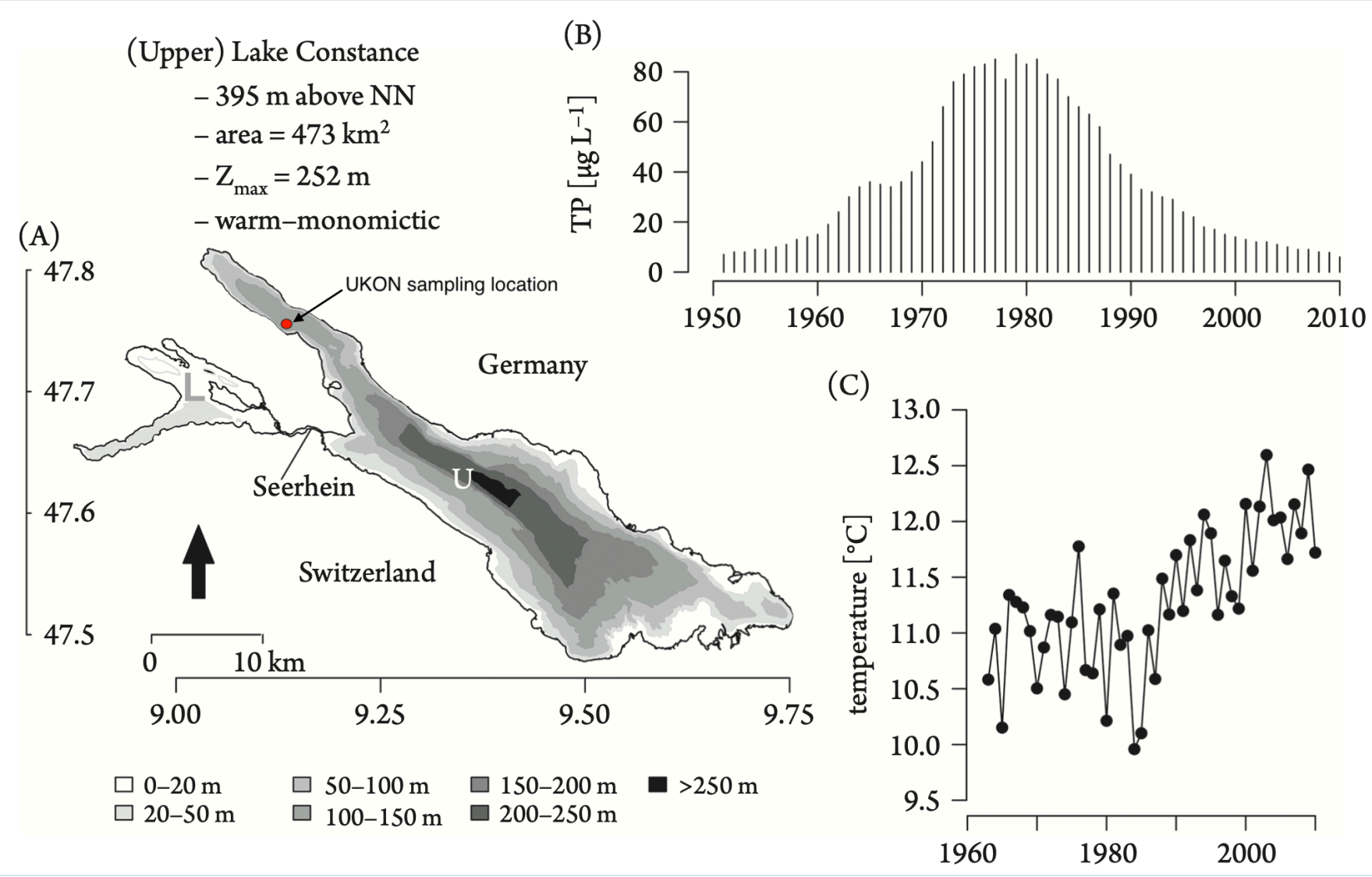
Plankton regime shifts can be detected and partly explained by biotic drivers. Whether they propagate and can be predicted is still open.



# Outline

1. What we know: background
2. Research questions
3. Methods: our pipeline
4. Results: detection of regime shifts
5. Explain regime shifts
6. Propagation of regime shifts
7. Conclusions

# What we know: background




**Lake Constance**
473 km², 252 m deep, warm-monomictic

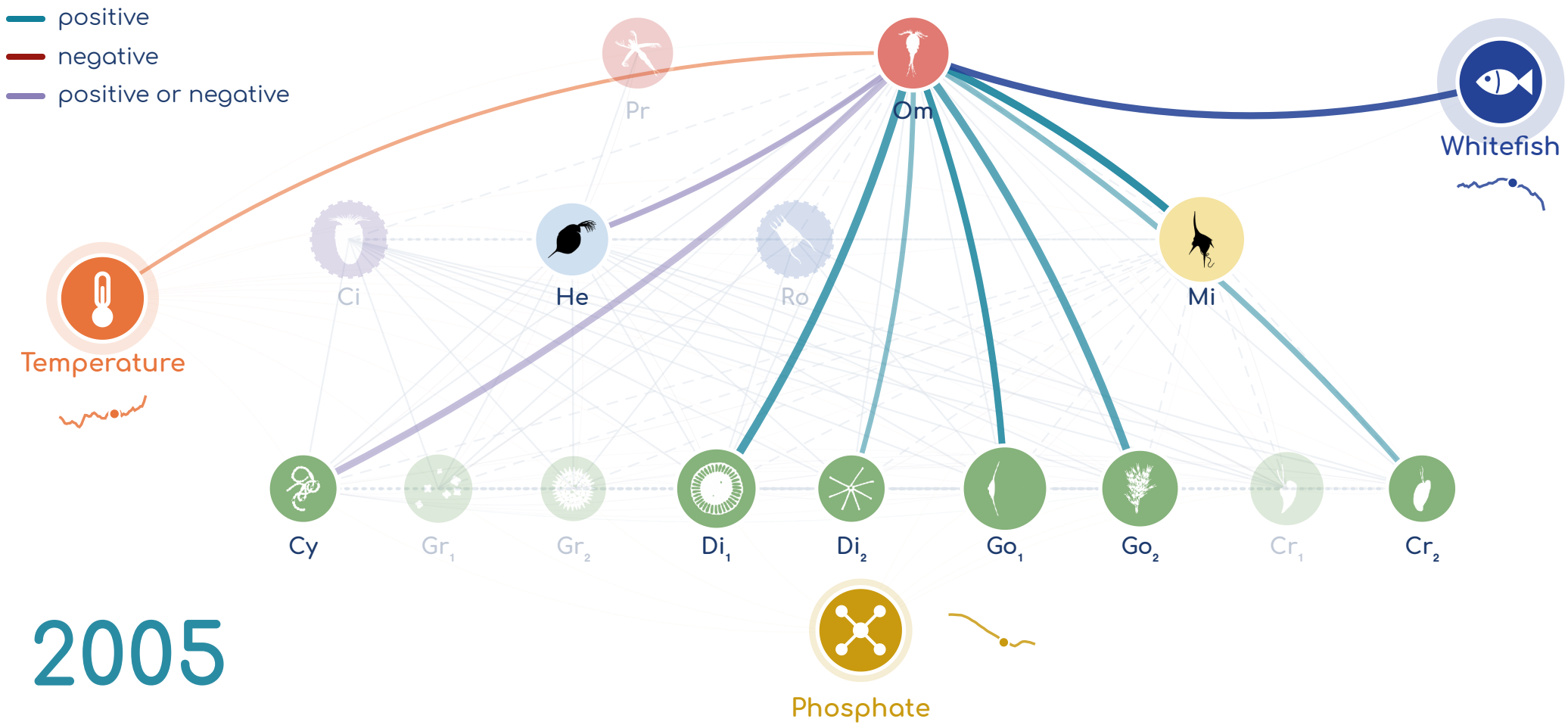
**Phosphate:** peak around 1980 (~85 µg/L), back to ~6 µg/L: **re-oligotrophication**

**Temperature:** warming of about 1 °C since the 1960s



Figure: Dietmar Straile, University of Konstanz (UKON sampling location, total phosphorus TP, annual temperature)

# What we know: the plankton community




Spotlight: what we expect to drive the omnivores (Om)



Network: Merz et al. 2023, Nat. Clim. Change · hypotheses: notebook 18 · Ci, Ro: not in the data

# Research questions



## 1 · Detect

Do plankton guilds show abrupt regime shifts?



## 2 · Explain

Which drivers explain a regime shift?



## 3 · Propagate

Did the shift of one guild cause the shift of another?



## 4 · Predict

Can we forecast regime shifts? (work in progress)

# Methods: our pipeline

1

## Data

Monthly IGKB monitoring, 1975–2022: 13 plankton guilds, temperature, phosphate, whitefish

2

## Detect

BEAST change point, Bayesian difference in means, abruptness > 1

3

## Explain

GP-EDM of each guild; hypotheses from the literature tested with heat maps and partial dependence

4

## Propagate

Shift-removal test: re-predict with 100 shift-free surrogates of the driver

5

## Predict

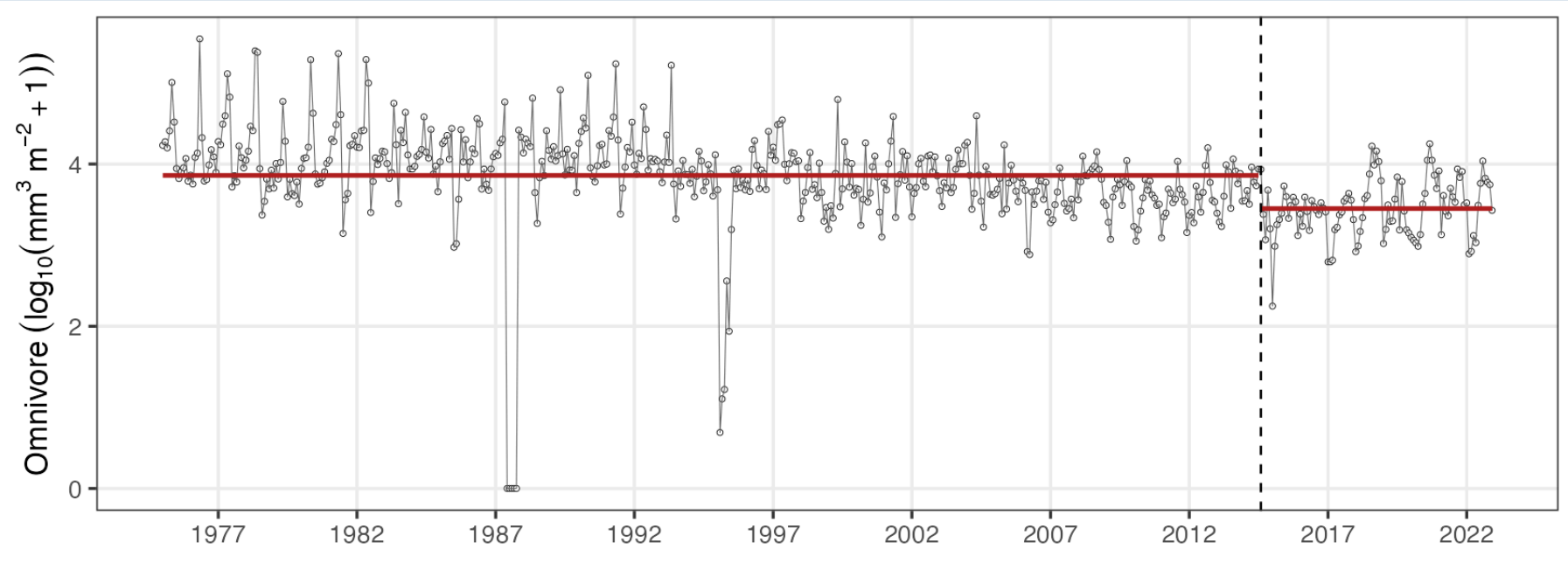
Forward validation of the models

WORK IN PROGRESS



Gilarranz et al. 2022 · Munch et al. 2017 (GPEDM) · Merz et al. 2023

# Results: detection of regime shifts




Omnivores (copepods), Lake Constance, monthly biovolume 1975–2022. Dashed: detected shift; red: mean of each period.



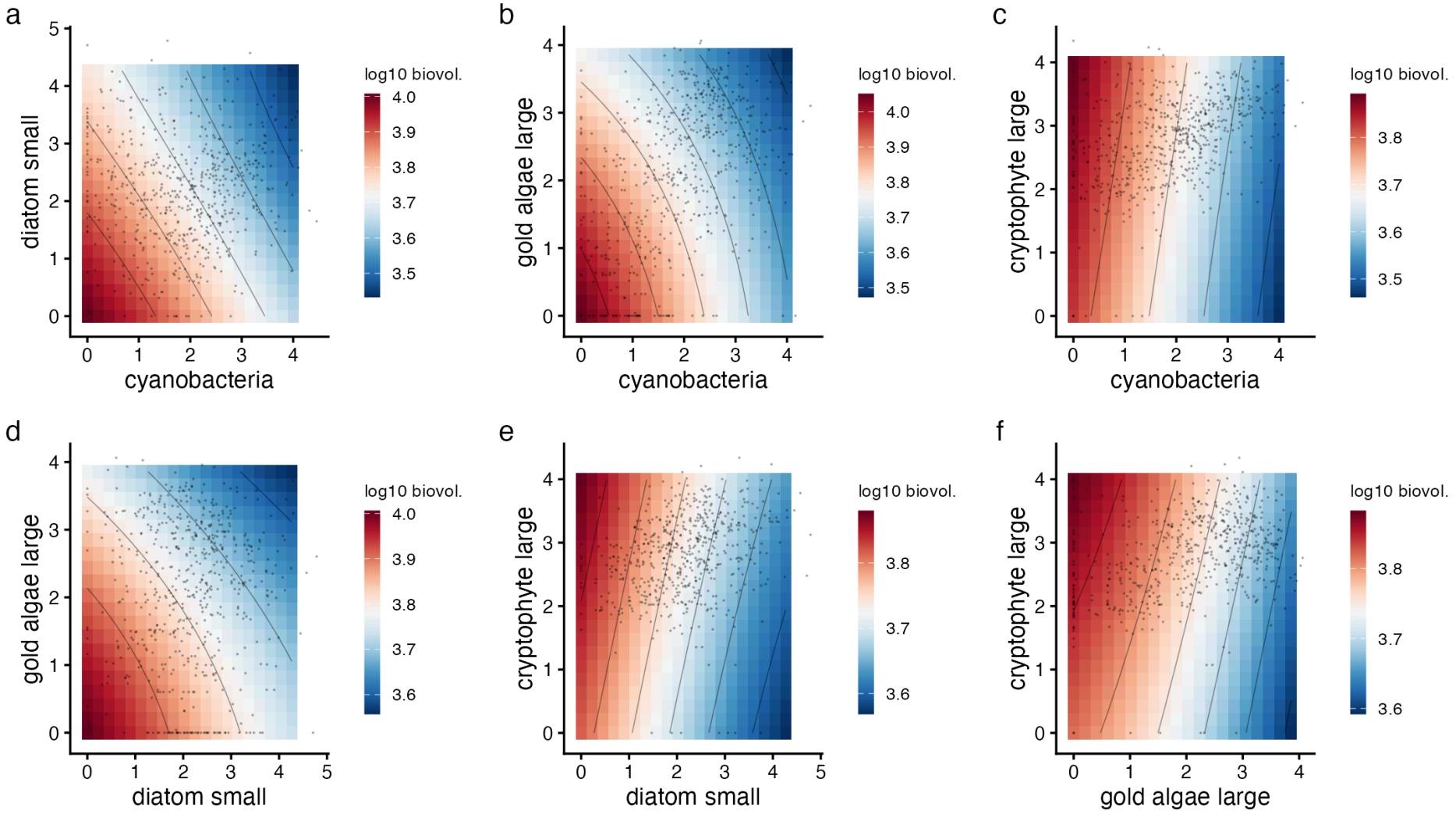
**Omnivores: shift in Aug 2014**

abruptness **1.19** · change-point prob. **0.87**

Also: small green algae, 1984 and 1993

**The omnivores dropped abruptly in 2014**

# Explain regime shifts





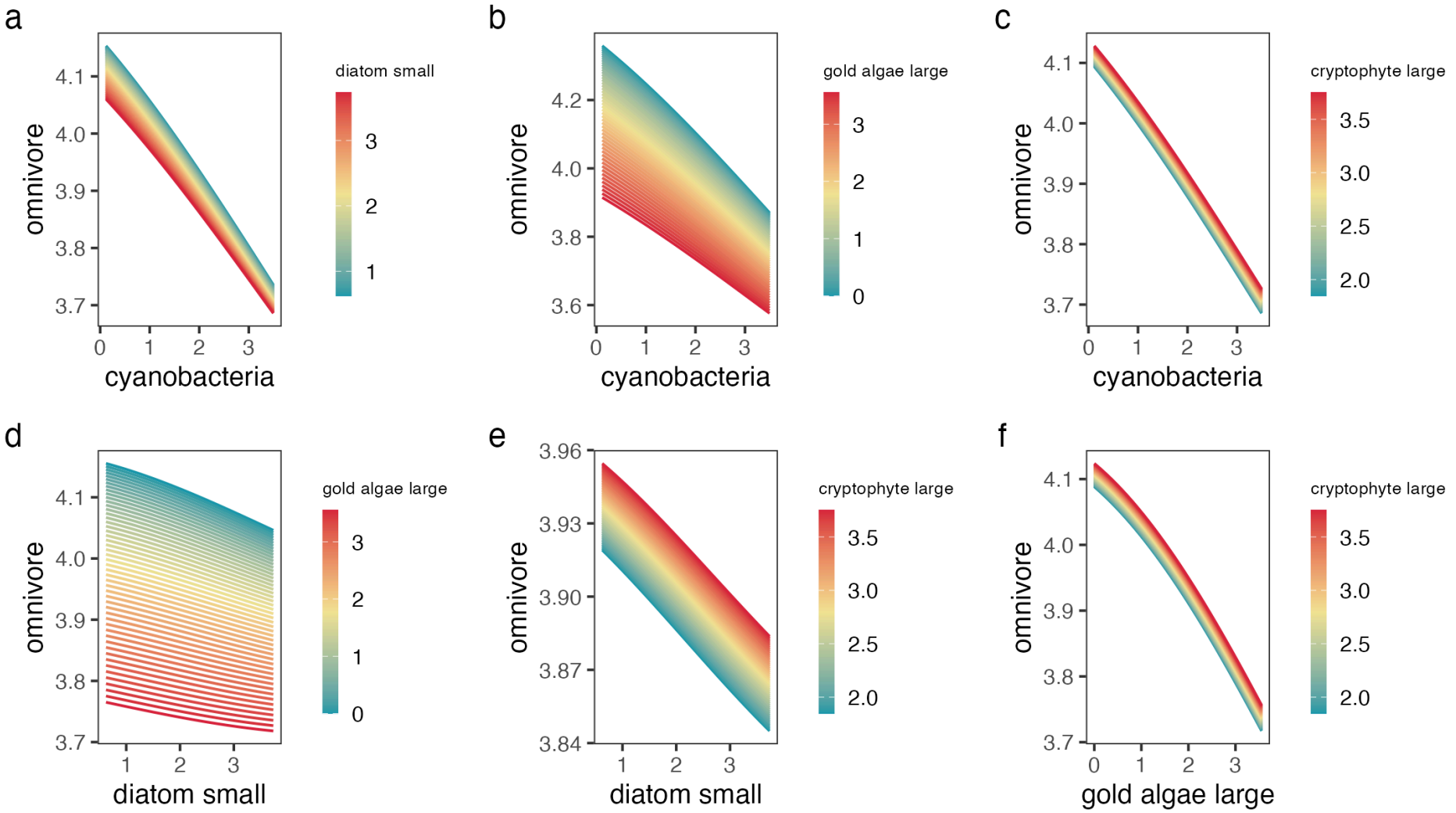

GP-EDM, LOO $R^2$ = **0.81**

Omnivores **decrease** with cyanobacteria (×2.2–2.4), small diatoms (×1.7–1.8), large gold algae (×1.6–1.7)

Prey hypothesis: more prey, more omnivores. **Not supported**

Heat maps: axes in log10(mm³ m⁻² + 1); dots: observed months

# Explain regime shifts: partial dependence






x axis: one predictor lines: levels of a second one

Cyanobacteria **lower** the omnivores at every level of the second predictor

Lines do not cross: no change of sign



Other predictors at their median; axes in log10(mm³ m⁻² + 1); lines: 5th to 95th percentile of the second predictor

# Propagation of regime shifts

**1 · Real predictor**




**2 · Shift-free surrogate (× 100)**




Gap between real and surrogate prediction = evidence for propagation

2 candidate pairs

**0** testable: the model must reproduce the shift

Both predictors shifted **after** the response

**No evidence of propagation yet: too few shifts**

# Conclusions



Shifts are detectable



Biotic drivers matter



Propagation, prediction: open

# Take home message

Plankton regime shifts can be detected and partly explained by biotic drivers. Whether they propagate and can be predicted is still open.



# Acknowledgments

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